

**Department of Electrical and Computer Engineering
National University of Singapore**

EE5508 Semiconductor Fundamentals (Semester I, AY2025/2026)

Assignment for Part II

A literature review of random lasers

Conduct a literature review on **random lasers** and submit a comprehensive report that demonstrates both depth of understanding and critical analysis. Your report should cover the fundamental principles underlying random laser operation, provide a concise historical overview of their development, discuss recent advancements and emerging trends, and identify current limitations and challenges in the field. To demonstrate originality, your report should include an independent component such as a comparative analysis of reported results based on different material systems or device structures, or any other elements you deem important for the development and applications of random lasers.

The report should be based on independent studies, not AI-generated summaries or synthesized content. AI tools may be used only for language refinement (e.g., improving grammar, clarity, or flow) or for creating illustrative elements (e.g., diagrams or schematics), and any such use must be clearly disclosed in the report.

Submit your report to Canvas under the folder 'Assignment report for part II' by 15 November 2025. The report should be written in Arial font, size 11, with single line spacing, and must not exceed 15 pages in total, including the cover page and references.

Please note that you are expected to submit a report based on your own efforts.
Plagiarism is strictly prohibited.